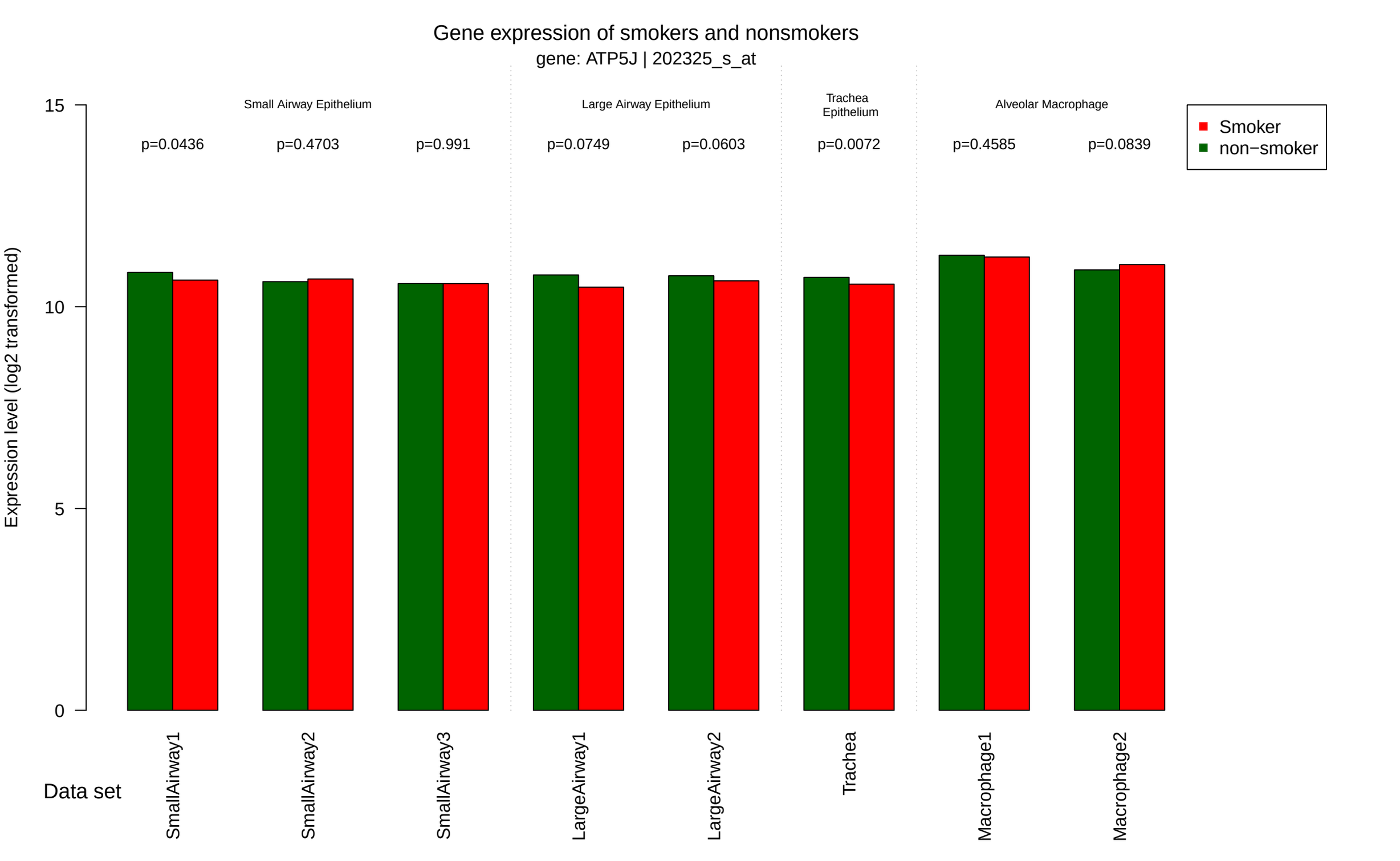
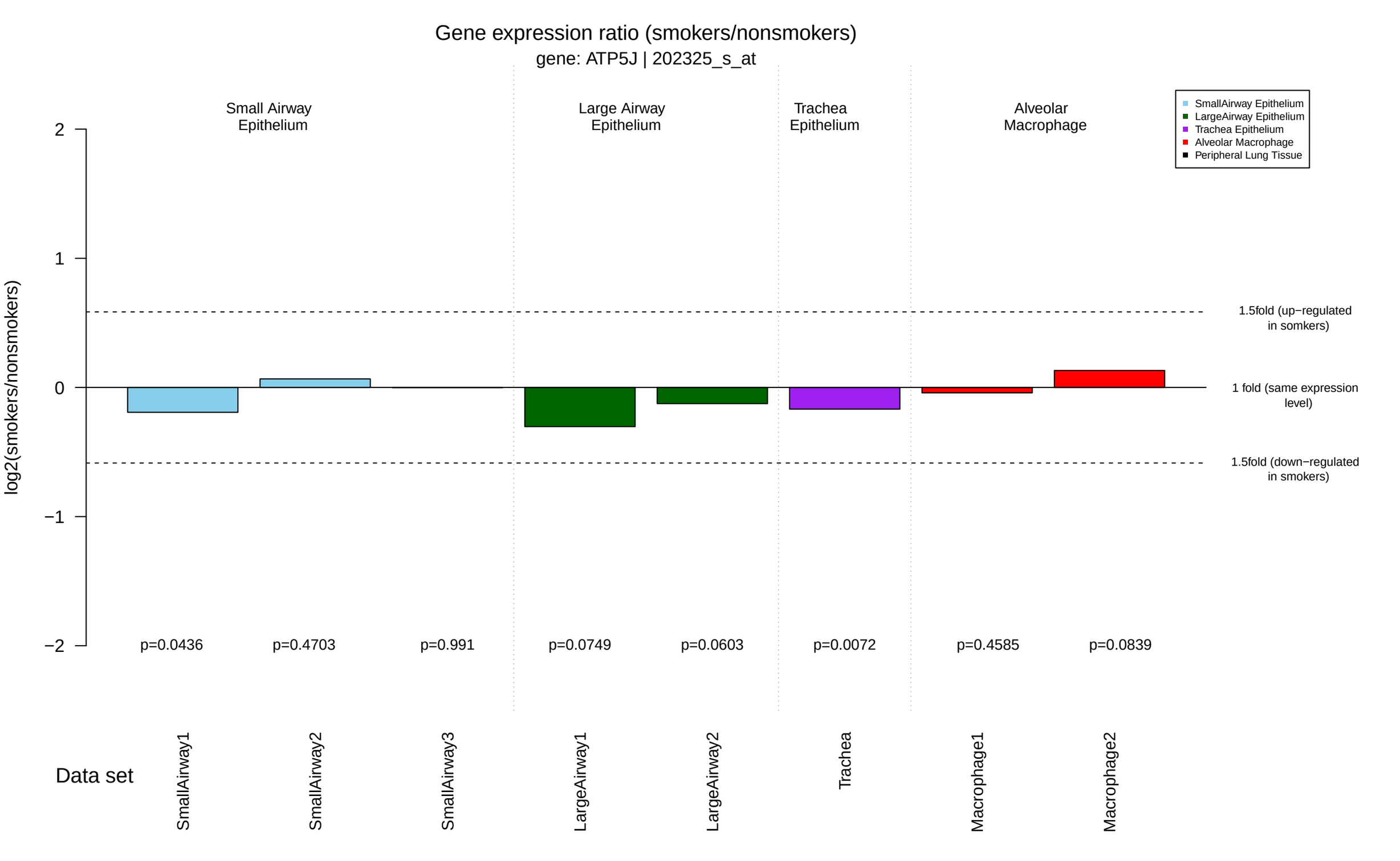
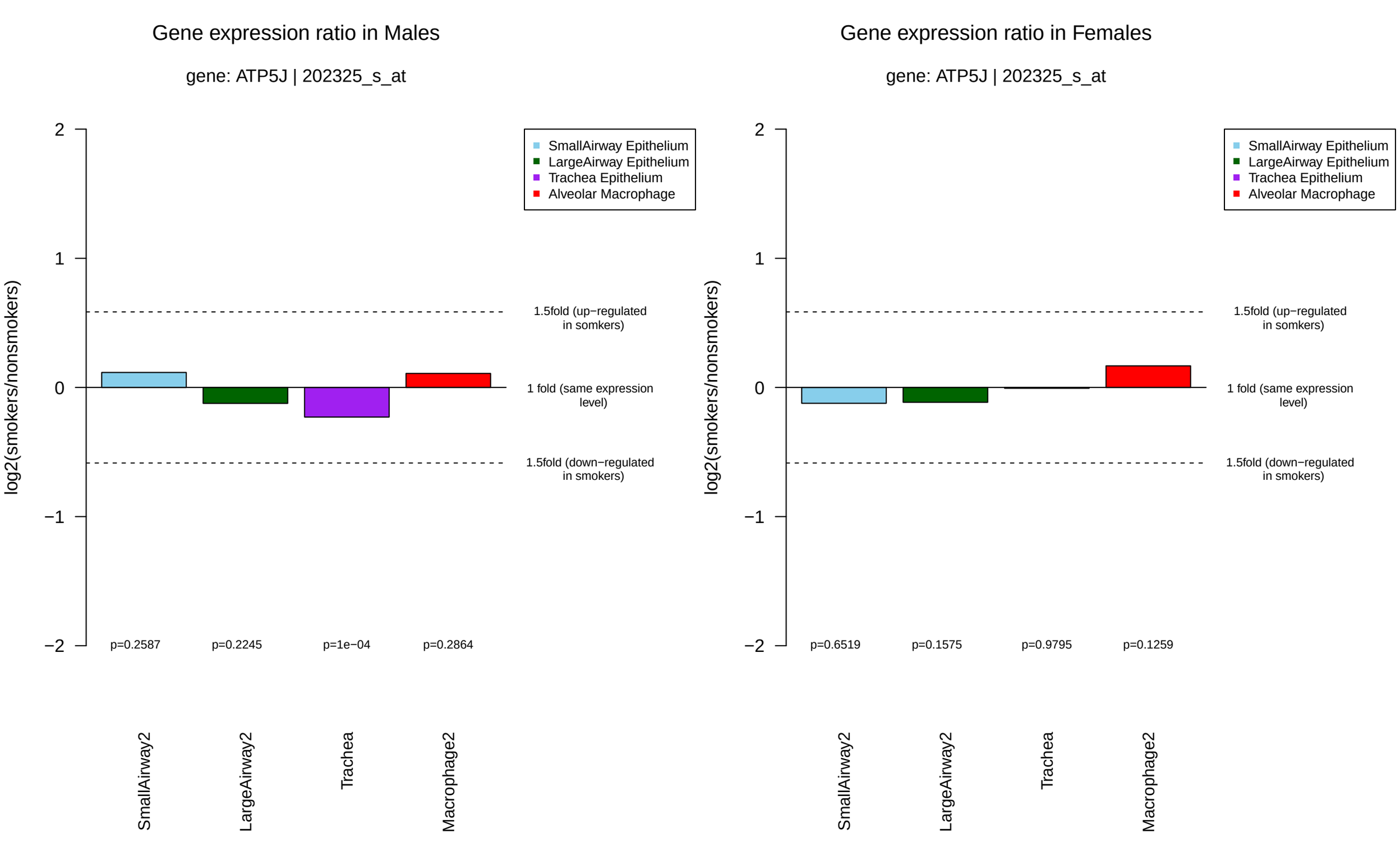
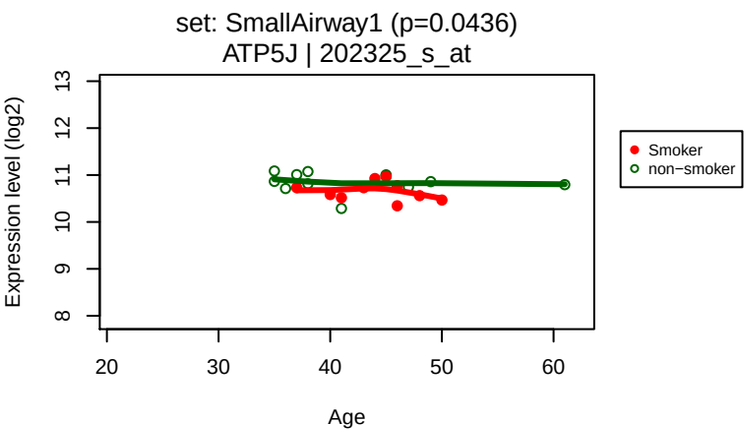
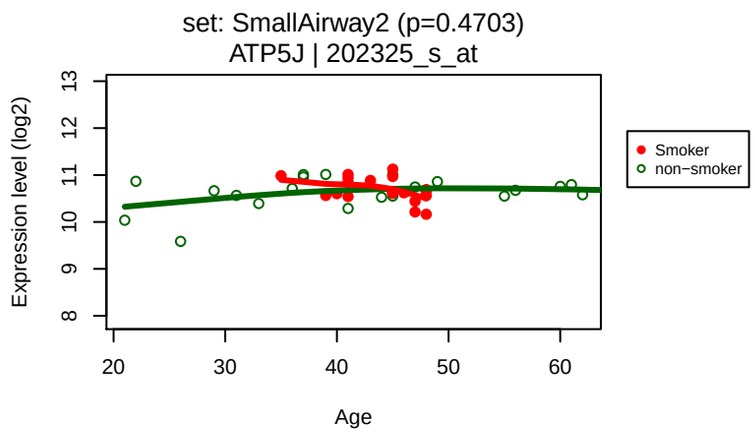
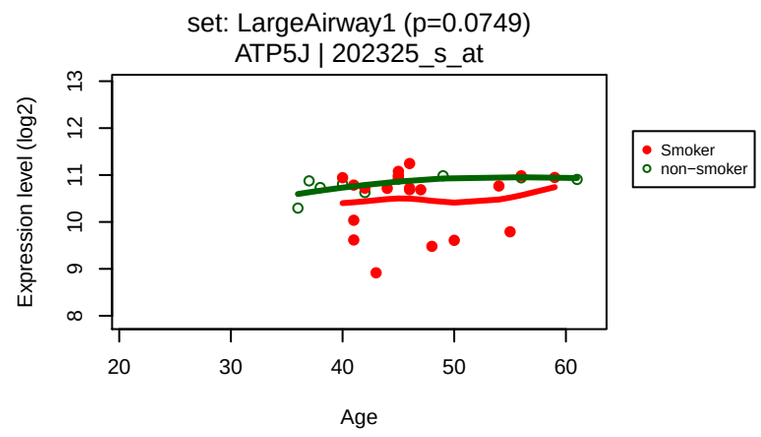

Gene expression of smokers and nonsmokers
gene: ATP5J | 202325_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0436
p=0.4703
p=0.991
p=0.0749
p=0.0603
p=0.0072
p=0.4585
p=0.0839
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

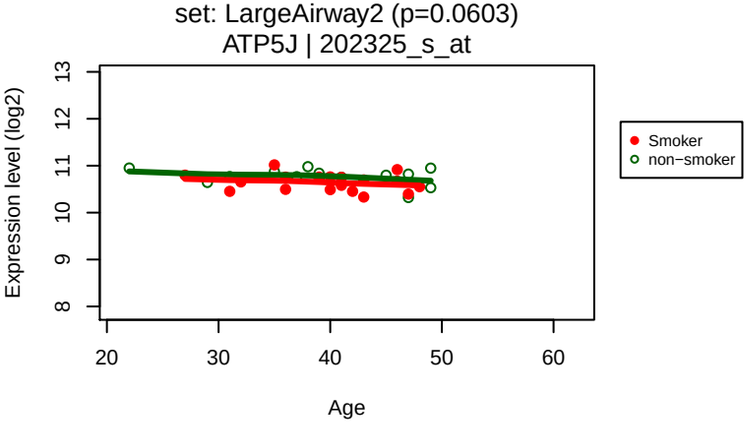

Gene expression ratio (smokers/nonsmokers)
gene: ATP5J | 202325_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0436
p=0.4703
p=0.991
p=0.0749
p=0.0603
p=0.0072
p=0.4585
p=0.0839
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

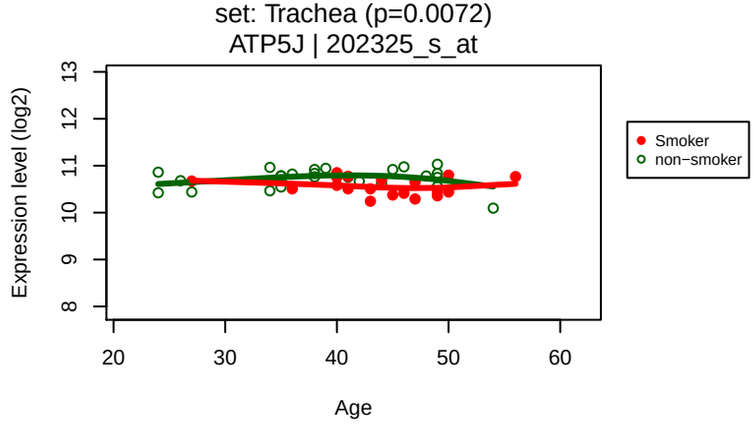

Gene expression ratio in Males
gene: ATP5J | 202325_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2587
p=0.2245
p=1e−04
p=0.2864
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATP5J | 202325_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6519
p=0.1575
p=0.9795
p=0.1259
SmallAirway2
LargeAirway2
Trachea
Macrophage2

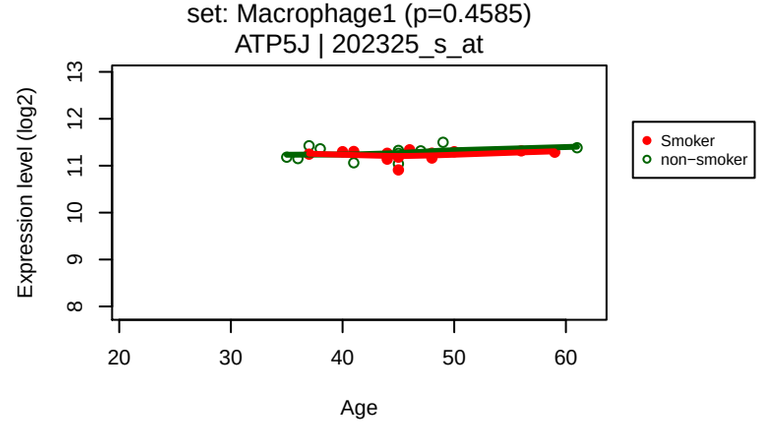

set: SmallAirway1 (p=0.0436)
ATP5J | 202325_s_at
Expression level (log2)
Age
Smoker
non−smoker

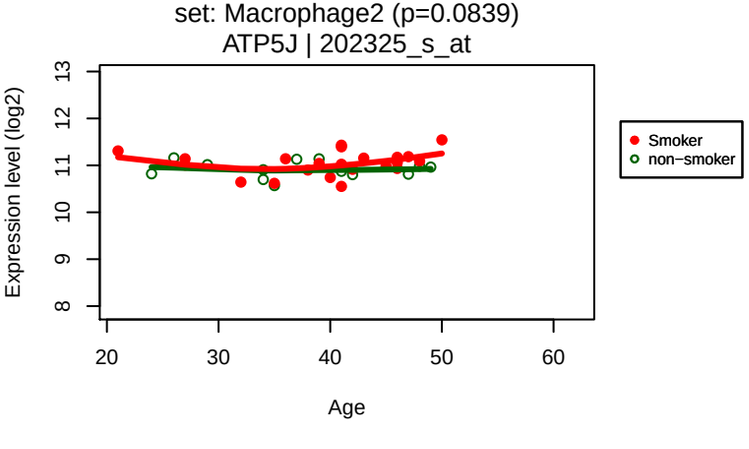

set: SmallAirway2 (p=0.4703)
ATP5J | 202325_s_at
Expression level (log2)
Age
Smoker
non−smoker

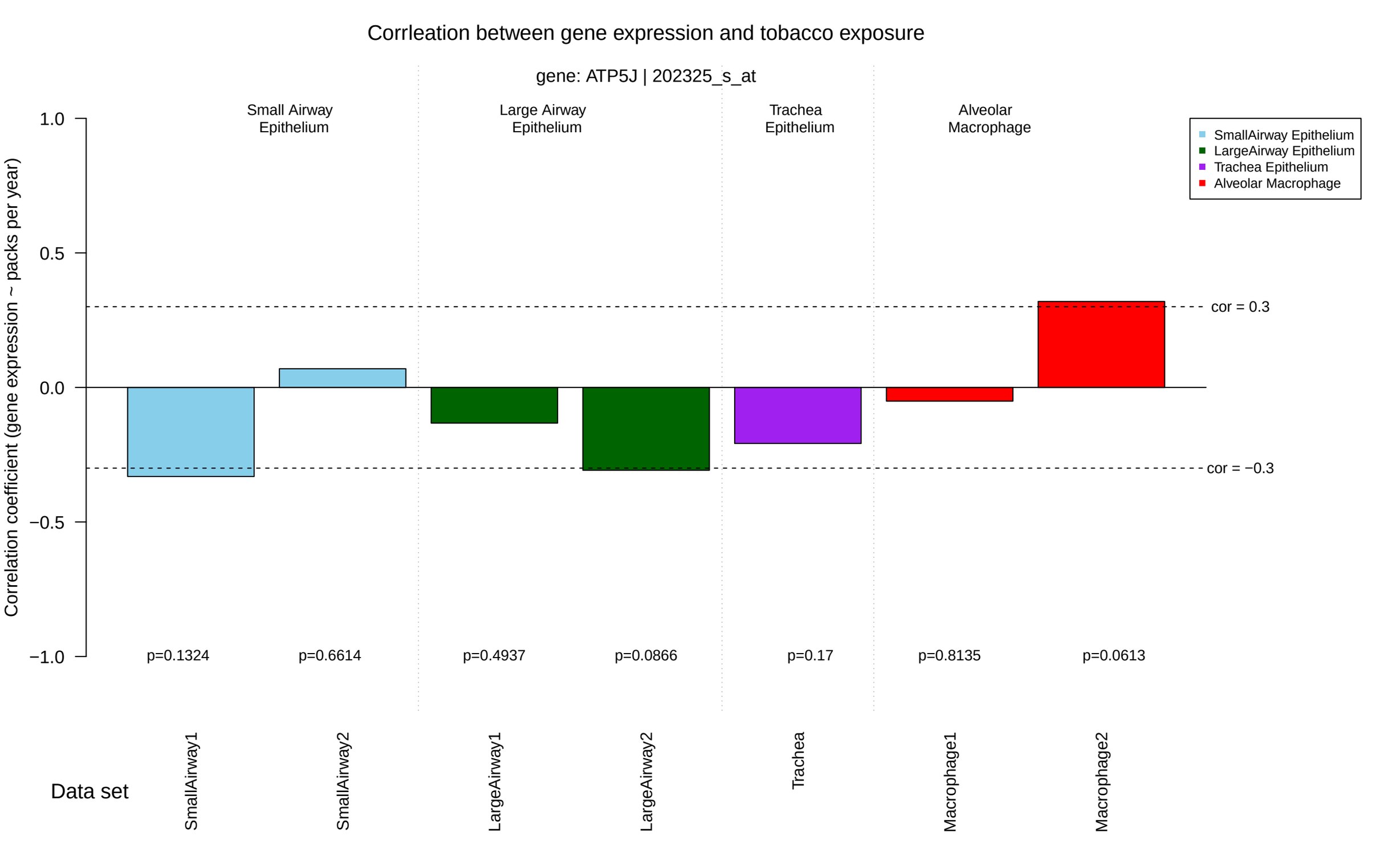

set: LargeAirway1 (p=0.0749)
ATP5J | 202325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0603)
ATP5J | 202325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0072)
ATP5J | 202325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4585)
ATP5J | 202325_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0839)
ATP5J | 202325_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATP5J | 202325_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1324
p=0.6614
p=0.4937
p=0.0866
p=0.17
p=0.8135
p=0.0613
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2